

Understanding Linear Regression

A Beginner's Guide: Finding the simple line in your data.

What is Linear Regression?

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Linear Regression is a way to find a straight line that best describes the relationship between two things (variables).

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Think of it as: Predicting one value based on another.

(e.g., Can we predict your test score based on how many hours you study?)

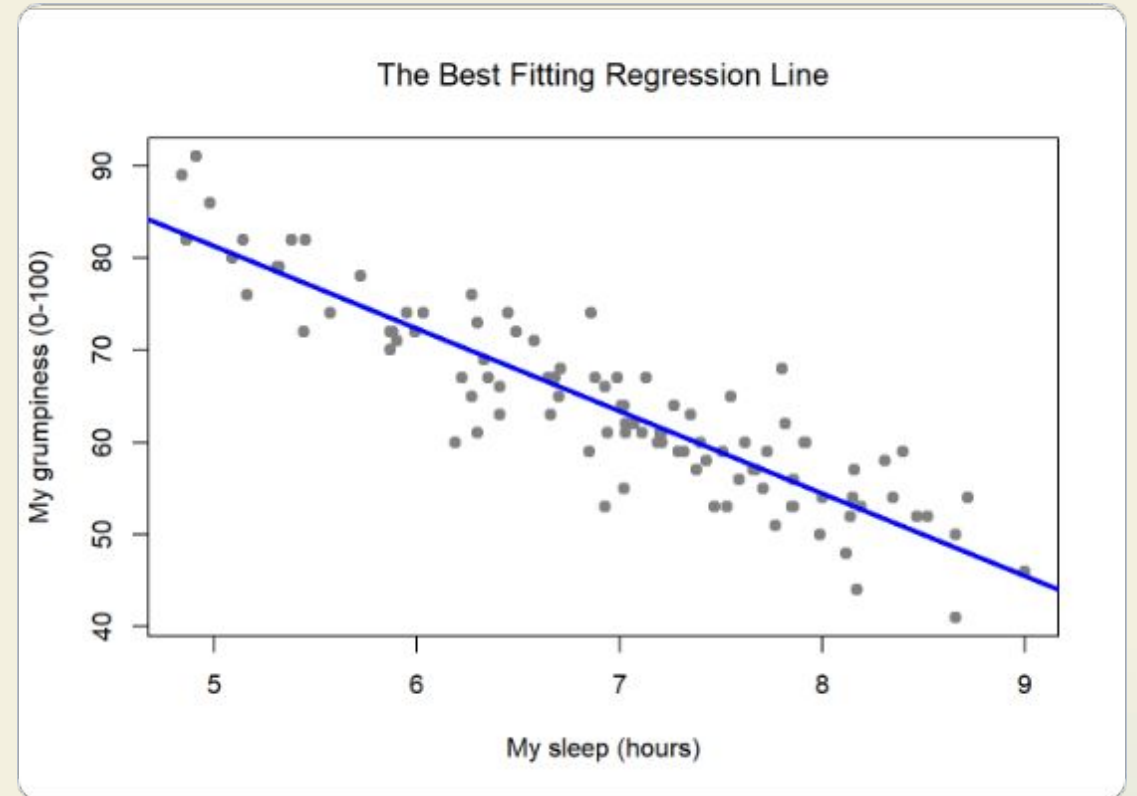
The Core Idea: The Best Fit Line

Finding the Pattern

Imagine you have a scatter plot of data points, like the one on the right. Each dot represents one piece of data (e.g., one student's hours and score).

Linear Regression isn't just connecting the dots. It's about drawing a **single straight line** that runs as close as possible to **all** the points at once.

This line represents the "best guess" or the underlying trend in the data.



A Simple Example: Study Hours vs. Test Score

Independent Variable (X)

This is the 'cause' or 'input'. It's the variable we think has an effect on the other.

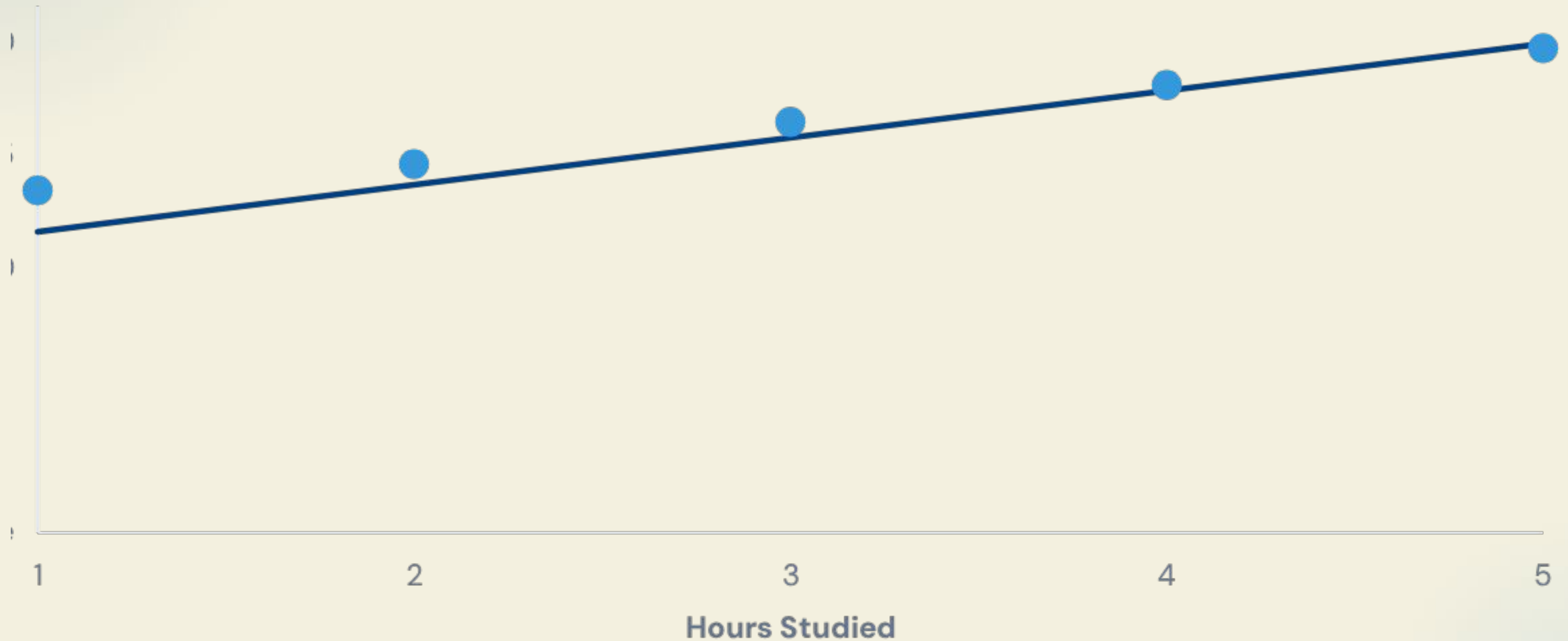
e.g., Hours Studied

Dependent Variable (Y)

This is the 'effect' or 'output'. It's the variable we are trying to predict.

**e.g., Test
Score**

Visualizing Our Example



The blue line is our 'linear regression model.' It's the best guess for the relationship between hours and scores.

The "Equation" (Don't Worry!)



Y

The value we want to predict
(Test Score).



X

The input value we know (Hours
Studied).



m & b

The 'slope' and 'intercept'. The
model **learns** these values to
find the best line.

How Does it "Learn"?

Goal:
Minimize Error

Ordinary Least Squares

The model finds the line that has the smallest **total error**.

The 'error' (or 'residual') is the distance from an actual data point to the line. The goal is to minimize the **sum of these squared errors**. This is why it's called "Ordinary Least Squares."

It's like finding the perfect "average" line that balances all the errors.

Example Code (Python)

Easy with Python

Using the `scikit-learn` library, you can train a model in just a few lines of code.

- **1. Import:** Bring in the `LinearRegression` model.
- **2. Prepare Data:** Create your sample data (X and Y).
- **3. Fit Model:** This is the 'learning' step where the model finds the best line.
- **4. Predict:** Use the trained model to predict new values.



The Code Itself (Python + scikit-learn)

```
from sklearn.linear_model import LinearRegression
import numpy as np

# 2. Prepare Data (X must be a 2D array)
X = np.array([[1], [2], [3], [4], [5]]) # Study Hours
y = np.array([65, 70, 78, 85, 92])      # Test Score

# 3. Create and Fit the model
model = LinearRegression()
model.fit(X, y)

# 4. Predict score for 3.5 hours
prediction = model.predict(np.array([[3.5]]))
# print(prediction) --> [81.5]
```

Another Example: Real Estate

Predicting House Prices

We can also predict a house's price based on its size.

- **Independent Variable (X):** Square Feet
- **Dependent Variable (Y):** Price

The model would find the best-fit line to describe how much the price typically increases for each additional square foot.

This helps you estimate a price for a new listing based on its size.



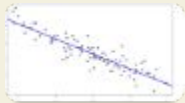
When to Use It (Assumptions)

-  **Linear Relationship:** The data should *actually* look like a straight line (not a U-shape or a curve).
-  **Independence:** Your data points aren't related to each other (e.g., one student's score doesn't affect another's).
-  **Normality:** The 'errors' (residuals) are "normally distributed" – they look like a bell curve around the line.
-  **Homoscedasticity:** The errors are spread out evenly along the line, not fanning out or bunching up.

Questions?

Thank you for your time.

Image Sources



https://learningstatisticswithr.com/book/lsr_files/figure-html/regression1a-1.png

Source: learningstatisticswithr.com



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